

MOHAMMED NAFEES H

Robotics & AI Engineering Student | ROS 2 • Robotics Controls • Raspberry Pi • Pi Camera • OpenCV

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PROFESSIONAL SUMMARY

Robotics & AI Engineering student at SASTRA University with hands-on experience from schooling through college in building robotics and embedded systems projects. Strong practical background in ROS 2, robotics controls, Raspberry Pi, Pi Camera, Ubuntu, OpenCV, and robot integration workflows. Earlier exposure includes Arduino-based prototyping during schooling. Interested in industrial robotics, autonomous systems, perception, navigation, and real-world deployment-oriented engineering.

EDUCATION

SASTRA Deemed University — B.Tech, Robotics & Artificial Intelligence
Current Student | Class of 2028 | Tamil Nadu, India

TECHNICAL SKILLS

- Robotics: ROS 2, robotics controls, robot integration, mobile robots, quadruped kinematics, sensor interfacing, control basics
- Programming: Python, C/C++, Linux shell scripting, Arduino basics
- Hardware: Raspberry Pi 5, Raspberry Pi Zero 2 W, Pi Camera, motor drivers, encoders, line sensors, sensor/actuator interfacing
- Software & Tools: Ubuntu 24.04, OpenCV, Git, VS Code, CoppeliaSim, Fusion 360, EasyEDA/KiCad basics
- Interests: Industrial robotics, autonomous navigation, robot perception, embedded AI, reinforcement learning

EXPERIENCE

Confidential Industrial Robotics Project | ROS 2, Robotics Controls, System Integration

Project details withheld due to signed confidentiality / NDA obligations

- Contributing to an industry-level robotics project involving ROS 2-based workflows, controls-oriented development, embedded integration, and system-level coordination.
- Supporting real-world engineering tasks across robotics software, hardware coordination, and practical deployment-focused development in a professional setting.
- Responsible for sensor and actuator identification, evaluation, and procurement coordination for a project hardware budget of approximately ₹10 lakh.
- Working with industrial-grade components and practical engineering constraints while maintaining strict confidentiality.

PROJECTS

QR-Based Mapping and Navigation Robot | Raspberry Pi 5, Pi Camera, OpenCV, Python

- Building a robot pipeline for QR-based cell identification, map generation, shortest-path planning, and dashboard-based monitoring using Raspberry Pi and camera-based perception.
- Designed with separate control, UI, firmware, and testing layers for easier debugging and future expansion.

Quadruped Robot Development | Kinematics, robotics controls, embedded integration

- Worked on inverse kinematics and motion logic for a quadruped robot, focusing on mathematically grounded leg movement.
- Explored how kinematics, control theory, and practical actuation connect in stable locomotion development.

Embedded Vision Modules for Robotics | Raspberry Pi, Pi Camera, OpenCV, Python

- Developing reusable camera-based perception modules that can be embedded into broader robotics systems rather than presented as isolated demo projects.
- Focused on practical integration of Raspberry Pi, camera input, and OpenCV-based processing for real robotics workflows.

Schooling-Era Robotics Projects | Arduino-based prototyping, sensors, motor drivers

- Completed numerous small robotics and embedded builds during schooling; Arduino was used mainly in this phase for early prototyping, control experiments, wiring, and debugging.

LEADERSHIP & INVOLVEMENT

- Member, Robotics Club — active in robotics learning, project building, experimentation, and applied technical development.

EXPERIENCE HIGHLIGHTS

- Strong practical learning style built around reverse-solving problems, debugging from failure states, and improving through implementation.
- Built and tested robotics systems across software, electronics, and mechanical integration instead of working only in simulation.
- Comfortable with Linux-based robotics environments, hardware bring-up, sensor debugging, rapid prototyping, and project execution under confidentiality constraints.